DEPARTEMENT STATISTIEK EN AKTUARIËLE WETENSKAP
DEPARTMENT OF STATISTICS AND ACTUARIAL SCIENCE

STATISTIEK EN DATAWETENSKAP 188
STATISTICS AND DATA SCIENCE 188

TUTORIAAL 8 / *TUTORIAL* 8

Die meeste probleme is aangepas uit BASIC Business Statistics – Berenson, Levine & Krebiel, 10th ed, Prentice Hall. / *Most problems are adapted from BASIC Business Statistics – Berenson, Levine & Krebiel, 10th ed, Prentice Hall*.

1. Indien $\bar{x} = 85$, $\sigma = 8$ en $n = 64$, stel 'n 95% vertrouensinterval vir die populasiegemiddelde μ op. / *If* $\bar{x} = 85$, $\sigma = 8$ *and* $n = 64$, *construct a* 95% *confidence interval for the population mean* μ.

2. Indien $\bar{x} = 125$, $\sigma = 24$ en $n = 36$, stel 'n 99% vertrouensinterval vir die populasiegemiddelde μ op. / *If* $\bar{x} = 125$, $\sigma = 24$ *and* $n = 36$, *construct a* 99% *confidence interval for the population mean* μ.

3. 'n Marknavorser beweer dat sy 95% vertroue het dat die gemiddelde maandelikse verkope van 'n produk tussen R170 000 en R200 000 is. Verduidelik die betekenis van hierdie stelling. / *A market researcher states that she has* 95% *confidence that the mean monthly sales of a product are between R*170 000 *and R*200 000. *Explain the meaning of this statement.*

4. Die inspeksie-afdeling van 'n koeldrankvervaardiger wil die werklike hoeveelheid gaskoeldrank in 2-liter bottels ondersoek. Die vervaardiger het dit aan die inspeksie-afdeling bekend gemaak dat die populasie standaardafwyking vir 2-liter bottels 0.05 liter is. 'n Ewekansige steekproef van 100 2-liter bottels by hierdie vervaardiger lewer 'n steekproefgemiddelde van 1.99 liter. / *The inspection division of a cooldrink manufacturer wants to estimate the actual amount of soft drink in 2-litre bottles at a local bottling plant. The bottling plant has informed the inspection division that the population standard deviation for 2-litre bottles is* 0.05 *litre. A random sample of* 100 *2-litre bottles at this bottling plant indicates a sample mean of* 1.99 *litres*.

   (a) Stel 'n 95% vertrouensinterval vir die populasiegemiddelde hoeveelheid gaskoeldrank in elke bottel op. / *Construct a* 95% *confidence interval for the population mean amount of soft-drink in each bottle*.
   (b) Is dit nodig om aan te neem dat die populasie van gaskoeldranke se inhoud normal verdeel is? Motiveer u antwoord. / *Must you assume that the population of soft-drink content is normally distributed? Motivate your answer*.

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(c) Verduidelik waarom ’n waarde van 2.02 liter vir ’n enkele bottel se inhoud nie ongewoon is nie, alhoewel dit buite die vertrouensinterval lê wat u bereken het. / *Explain why a value of* 2.02 *litres for the content of a single bottle is not unusual, even though it is outside the confidence interval you calculated*.

(d) Gestel dat die steekproefgemiddelde 1.97 liter was. Hoe sou dit u antwoord in (a) verander? / *Suppose that the sample mean had been* 1.97 *litres. How would this change your answer to (a)?*

5. Indien u 95% vertroue wil hê in die beraming van die populasiegemiddelde binne ’n steekproeffout van ± 5 eenhede en kan aanneem dat die standaardafwyking 15 is, wat is die steekproefgrootte wat benodig word? / *If you want to be* 95*% confident of estimating the population mean to within a sampling error of* ±5 *units and the standard deviation is assumed to be* 15*, what sample size is required*?

6. Indien u 99% vertroue wil hê in die beraming van die populasiegemiddelde binne ’n steekproeffout van ± 20 en kan aanneem dat die standaardafwyking 100 is, wat is die steekproefgrootte wat benodig word? / *If you want to be* 99*% confident of estimating the population mean to within a sampling error of* ±20 *and the standard deviation is assumed to be* 100*, what sample size is required*?

7. In 2001 is beraam dat 45% van Suid-Afrikaanse huishoudings kruideniersware by geriefswinkels koop, en dat 29% kruideniersware by groothandelaars koop. Beskou ’n opvolgstudie wat op die laaste kalenderjaar fokus. / *In 2001 it was estimated that* 45*% of South African households purchased groceries at convenience stores and* 29*% purchased groceries at wholesalers. Consider a follow-up study focusing on the last calendar year*.

   (a) Watter steekproefgrootte word benodig om die populasieproporsie Suid-Afrikaanse huishoudings wat kruideniersware by geriefswinkels koop binne ± 0.02 met 95% vertroue te beraam? / *What sample size is needed to estimate the population proportion of South African households purchasing groceries at convenience stores to within* ±0.02 *with* 95*% confidence*?

   (b) Watter steekproefgrootte word benodig om die populasieproporsie Suid-Afrikaanse huishoudings wat kruideniersware by groothandelaars koop binne ± 0.02 met 95% vertroue te beraam? / *What sample size is needed to estimate the population proportion of South African households purchasing groceries at wholesalers to within* ±0.02 *with* 95*% confidence*?

8. In ’n opname van 326 sakelui het 303 geantwoord dat hul voorleggings al deur die lui van ’n selfoon onderbreek is, en 23 het geantwoord dat dit nog nie met hul gebeur het nie (“You Say”, Presentations: *Technology and Techniques for Effective Communication*, January 2003, 18). / *In a survey of* 326 *business people,* 303 *answered that their presentations have been interrupted by a ringing cell phone, and* 23 *answered that this has not yet happened to them* (“You Say”, Presentations: *Technology and Techniques for Effective Communication*, January 2003, 18).

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(a) Stel ’n 95% vertrouensinterval vir die populasieproporsie sakelui wie se voorleggings al deur selfone onderbreek is, op. / *Construct a 95% confidence interval for the population proportion of business people whose presentation have been disturbed by cell phones*.
(b) Interpreteer die vertrouensinterval in (a). / *Interpret the confidence interval in (a).*
(c) Hoe groot steekproef word benodig om ’n opvolgstudie te loods wat 95% vertroue sal lewer dat die puntberamer binne ±0.04 van die populasieproporsie is? / *To conduct a follow-up study that would provide 95% confidence that the point estimate is correct to within ±0.04 of the population proportion, how large a sample is required?*
(d) Hoe groot steekproef word benodig om ’n opvolgstudie te loods wat 99% vertroue sal lewer dat die puntberamer binne ±0.04 van die populasie proporsie is? / *To conduct a follow-up study that would provide 99% confidence that the point estimate is correct to within* ±0.04 *of the population proportion, how large a sample is required?*

9. Bepaal die kritieke waarde van t in elk van die volgende omstandighede. / *Determine the critical value of t in each of the following circumstances*.

(a) $1 - \alpha = 0.95, n = 10$
(b) $1 - \alpha = 0.99, n = 10$
(c) $1 - \alpha = 0.95, n = 32$
(d) $1 - \alpha = 0.95, n = 65$
(e) $1 - \alpha = 0.90, n = 16$

10. Regulering vereis dat bandevervaardigers prestasie-inligting aan die kant van ’n band moet verskaf. Een belangrike maatstaf vir die prestasie van die band is die loopvlak slytasie-indeks, wat die weerstand teen slytasie toon in vergelyking met ’n basis van 100. ’n Band met ’n indeks van 200 moet gemiddeld twee keer so lank duur as ’n band van 100. Ons wil die werklike loopvlak slytasie-indeks van ’n handelsmerk skat wat ’n indeks van 150 aandui. ’n Steekproef van $n = 24$ lewer ’n gemiddelde slytasie-indeks van 141.3 en ’n standaardafwyking van 19.1. */ Regulation requires tyre manufacturers to provide performance information on the side of a tyre. One important measure of tyre performance is the tread wear index, which shows resistance to wear compared with a base of 100. A tyre with an index of 200 should last twice as long, on average as a tyre graded as 100. We want to estimate the actual tread wear index of a brand that claims an index of 150. A sample of* $n = 24$ *indicates a mean wear index of 141.3 and a sample standard deviation of 19.1.*

(a) Neem aan dat die populasie van loopvlak slytasie-indekse normaal verdeel is, konstrueer ’n 90% vertrouensinterval vir die populgemiddelde slytasie-indeks vir bande wat deur hierdie vervaardiger onder hierdie handelsnaam vervaardig word. */ Assuming that the population of tread wear indexes is normally distributed, construct a 90% confidence interval for the population mean tread wear index for tyres produced by this manufacturer under this brand name.*

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(b) Dink u dat die verbruikersorganisasie die vervaardiger moet beskuldig dat hy bande vervaardig wat nie aan die prestasie-inligting aan die kant van die band voldoen nie? Verduidelik. / *Do you think that the consumer organization should accuse the manufacturer of producing tyres that do not meet the performance information provided on the side of the tyre? Explain.*

(c) Verduidelik waarom 'n waargenome loopvlak slytasie-indeks van 152 vir 'n bepaalde band nie ongewoon is nie, alhoewel dit buite die vertrouensinterval wat in (a) ontwikkel is, is. / *Explain why an observed tread wear index of 152 for a particular tyre is not unusual, even though it is outside the confidence interval developed in (a).*

11. Indien $\bar{x} = 75,\ s = 24,\ n = 36,\ en\ N = 200,$ konstrueer a 95% vertrouensinterval beraming van die populasie-gemiddeld, μ, as die steekproefneming gedoen is sonder terugplasing. / *If* $\bar{x} = 75,\ s = 24, n = 36,\ and\ N = 200,$ *construct a 95% confidence interval estimate of the population mean,* μ*, if sampling is done without replacement.*

12. Die bestuurder van 'n bank met 1 000 deposante wil die proporsie van sy deposante met meer as een rekening by die bank beraam. 'n Ewekansige steekproef van 100 deposante word sonder terugplasing gekies, en 30 verklaar dat hulle meer as een rekening by die bank het. / *The manager of a bank that has 1 000 depositors wants you to estimate the proportion of its depositors with more than one account at the bank. A random sample of 100 depositors is selected without replacement, and 30 state that they have more than one account at the bank.*

    (a) Construeer 'n 90% vertrouensinterval op van die populasieproporsie van die bank se deposante wat meer as een rekening by die bank het. / *Construct a 90% confidence interval for the population proportion of the bank's depositors who have more than one account at the bank.*

    (b) Watter steekproefgrootte is nodig om die populasieproporsie binne ± 0.05 met 90% vertroue te skat? / *What sample size is needed to estimate the population proportion to within* ± 0.05 *with 90% confidence?*

    (c) Wat is u antwoorde op (a) en (b) as die bank 2 000 deposante het? / *What are your answers to (a) and (b) if the bank has 2,000 depositors?*

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